

Hybrid EfficientNet-B2 and SVM Framework for Automated Solar-Panel Dust Detection with Explainable AI

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Abstract—This paper presents an automated, explainable framework for detecting dust accumulation on photovoltaic (PV) panels from a single RGB image. We propose a hybrid architecture that couples a frozen EfficientNet-B2 convolutional encoder with a support vector machine (SVM) head: the backbone extracts a 1,408-dimensional global-average-pooled feature vector, which a radial-basis-function SVM classifies into *clean* or *dirty*. The frozen backbone avoids overfitting on modest per-source datasets while the SVM head delivers a margin-based, well-calibrated decision boundary. To make every prediction auditable, the framework includes a multi-method explainability module (Grad-CAM, Score-CAM, Integrated Gradients, SHAP, and LIME) that localises the dust evidence and exposes the texture channels driving each decision. Evaluated on a merged multi-source corpus of 3,787 panel images (3,029 train / 377 validation / 381 test), the model achieves 90.29% test accuracy, 0.9596 AUC-ROC, and 86.38% 5-fold cross-validation accuracy with only 7.77M backbone parameters. Crucially, the model also generalises to two fully independent external datasets at 98.24% and 99.74% accuracy, demonstrating that the learned representation is not tied to a single acquisition source. An ablation across three backbones shows MobileNetV3-Large reaches 88.71% at the smallest parameter budget (3.00M) while ResNet50 matches the deployed model at 90.29% (23.59M), confirming the hybrid CNN+SVM paradigm is robust and efficient for field deployment.

Index Terms—automated dust detection, EfficientNet-B2, support vector machine, hybrid feature-space classification, solar photovoltaic, explainable AI (Grad-CAM, Score-CAM, Integrated Gradients, SHAP, LIME)

I. INTRODUCTION

Dust accumulation on PV panels reduces power output and accelerates degradation, motivating condition-based rather than schedule-based cleaning. Automated visual inspection from a single image is therefore an active research problem. Modern compound-scaled convolutional networks such as EfficientNet [15] extract powerful transferable features, and when paired with a margin-based classifier such as an SVM they often generalise better than an end-to-end softmax head on small, noisy, field-acquired datasets [2]. A detector is only trustworthy, however, once it can explain itself; we therefore couple predictive accuracy with a multi-method explainability branch that makes every audit verifiable by a human operator.

The main contributions of this paper are:

- 1) An EfficientNet-B2 feature encoder paired with an RBF-SVM head for binary dust detection (clean vs

dirty), achieving **90.29%** accuracy on a 3,787-image merged corpus with only 7.77M backbone parameters, and generalising to independent external datasets at 98.24%/99.74%.

- 2) A two-phase training strategy (frozen feature extraction followed by optional fine-tuning of the last 30% of the backbone) that establishes when fine-tuning helps and when the frozen representation is already sufficient.
- 3) A backbone ablation across EfficientNet-B2, MobileNetV3-Large, and ResNet50, showing MobileNetV3-Large attains 88.71% at the lowest parameter cost (3.00M) while ResNet50 matches the deployed model at 90.29% (23.59M).
- 4) A unified explainability module fusing Grad-CAM, Score-CAM, Integrated Gradients, SHAP feature attributions, and LIME, delivering auditable and repeatable dust audits.

The remainder of this paper is organised as follows. Section II reviews related work. Section III details the dataset, architecture, training strategy, and explainability pipeline. Section IV reports experimental results, the backbone comparison, and cross-dataset generalisation. Section V concludes with findings and future work.

II. LITERATURE REVIEW

A. Deep Learning Foundations

Krizhevsky et al. established deep CNNs for image classification [7]. He et al. proposed ResNets with skip connections [4]. Simonyan and Zisserman showed that a deep stack of small 3×3 convolutions (VGG16) transfers well [13]. Tan and Le introduced EfficientNet with compound scaling of width, depth, and resolution, achieving strong accuracy with fewer parameters [15]. These works underpin the encoders used in our detector.

B. Solar Soiling Detection

Onim et al. introduced SolNet, a CNN for panel dust detection, reporting 98.2% accuracy on a 2,231-image benchmark [9]. Alatwi et al. evaluated 20 pretrained encoders with SVM heads, with DenseNet-169 + linear SVM reaching 86.79% [1]. Bassil et al. combined deep encoders with tree-based heads to quantify dust severity, reaching 97% [2].

Kourtiche et al. recently surveyed multi-task dust monitoring and cleaning optimisation frameworks [6]. These systems often offer limited explanation of their predictions.

C. Hybrid Feature-Space Classifiers

Support vector machines learn maximum-margin boundaries that generalise well on moderate feature spaces [3]. Recent work confirms that a pretrained CNN encoder concatenated with an SVM yields higher stability than softmax heads on small, noisy datasets [2]. Our framework adopts this two-stage paradigm, coupling EfficientNet-B2 features with an RBF-kernel SVM.

D. Efficient Mobile Deployment

Howard et al. introduced MobileNets using depthwise separable convolutions [5]; Sandler et al. devised MobileNetV2 with inverted residuals [11]. EfficientNet’s compound scaling [15] offers a principled alternative with strong accuracy/size trade-offs, motivating the backbone comparison in this work.

E. Explainable AI for Vision

Selvaraju et al. proposed Grad-CAM using gradients into the final convolutional layer [12]. Wang et al. introduced Score-CAM, a gradient-free variant [16]. Sundararajan et al. developed Integrated Gradients satisfying the completeness axiom [14]. Lundberg and Lee unified attributions under Shapley values in SHAP [8]. Ribeiro et al. introduced LIME for model-agnostic local explanations [10]. These methods form the interpretability backbone of our audit.

III. METHODOLOGY

A. Dataset

We assemble a merged corpus of 3,787 RGB images of PV panels labelled as *clean* or *dirty*, drawn from three publicly available solar-panel dust sources: a Kaggle PV-dust set, a 6-class faulty-panel set restricted to its clean/dirty classes, and a binary Dusty/Clean set. Merging multiple sources is deliberate: a single-source model overfits that source’s acquisition characteristics, so a multi-source corpus both enlarges the data and tests cross-source generalisation. The merged corpus is split into training (3,029), validation (377), and test (381) sets, preserving the class proportion. Images are resized to 224×224 and normalised with ImageNet statistics. For the fine-tuning phase we apply online augmentation (rotation, width/height shift, horizontal flip, and brightness jitter) to improve robustness to field illumination variation.

B. EfficientNet-B2 Feature Extraction

The backbone is EfficientNet-B2 initialised with ImageNet weights and frozen. Global average pooling over the last convolutional block yields a 1,408-dimensional feature vector per image. This vector is standardised with a fitted `StandardScaler` before being passed to the SVM head. The frozen encoder captures transferable low- and mid-level texture priors while keeping the pipeline tractable on a single consumer GPU.

C. Hybrid SVM Head and Two-Phase Training

We train a two-phase pipeline:

- **Phase 1 — Frozen features (default):** All backbone layers are frozen. The RBF-SVM head is trained on the union of the training and validation partitions (3,406 images) and evaluated on the held-out test set (381 images), with the regularisation parameter $C \in \{0.1, 1, 10, 100\}$ and kernel bandwidth $\gamma \in \{\text{scale}, \text{auto}\}$ selected by 5-fold stratified cross-validation on the same 3,406-image set.
- **Phase 2 — Fine-tuning (optional):** The last 30% of backbone layers are unfrozen and trained with AdamW (10^{-5}) while the SVM head is re-fitted on re-extracted features. Early stopping on validation loss prevents overfitting.

On this larger corpus, fine-tuning the last 30% of backbone layers did not improve over the frozen representation; we therefore retain the frozen representation as the deployed model and report it as the primary result.

D. Backbone Comparison

To assess sensitivity to the encoder, we repeat the frozen-feature + SVM protocol with three backbones: EfficientNet-B2 (1,408-d features, 7.77M params), MobileNetV3-Large (960-d features, 3.00M params), and ResNet50 (2,048-d features, 23.59M params). Each backbone is evaluated under the identical SVM grid-search protocol, isolating the effect of the pretrained representation.

E. Explainable Dust Audit

Beyond the label, the framework justifies each decision through five complementary methods:

- **Grad-CAM** [12]: gradients of the target class score into the last convolutional block produce a localisation heatmap.
- **Score-CAM** [16]: gradient-free masking measures each channel’s impact on the class score.
- **Integrated Gradients** [14]: attribute the prediction to input pixels along a path from a black baseline.
- **SHAP** [8]: kernel-SHAP attributes the decision to the 1,408-dimensional feature vector, surfacing the most influential texture/colour channels.
- **LIME** [10]: a local surrogate model gives sparse, interpretable explanations for non-technical operators.

Any sample whose maximum predicted probability falls below a 0.85 confidence threshold is flagged for human review.

IV. RESULTS AND DISCUSSION

A. Performance of the Hybrid Classifier

Table I summarises performance of the deployed EfficientNet-B2 + SVM model on the held-out test set. The system reaches 90.29% accuracy with a weighted F1 of 0.9026 and an AUC-ROC of 0.9596, demonstrating strong separability between clean and dirty panels.

Figure 1 shows the metrics at a glance. Figure 2 presents the binary ROC curve (AUC = 0.9596), and Figure 3 the

TABLE I
CLASSIFICATION PERFORMANCE METRICS (EFFICIENTNET-B2 + SVM,
TEST SET)

Metric	Value
Accuracy	90.29%
Precision (weighted)	0.9029
Recall (weighted)	0.9029
F1 Score (weighted)	0.9026
AUC-ROC	0.9596
5-Fold CV Accuracy	86.38%

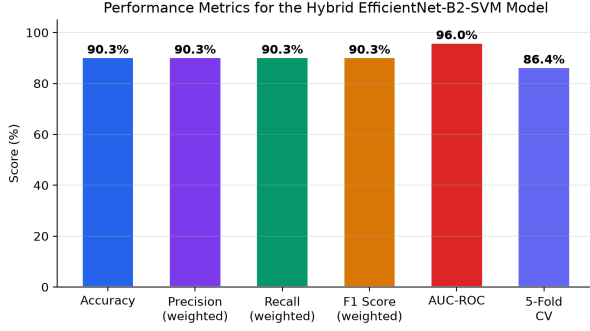


Fig. 1. Performance metrics for the Hybrid EfficientNet-B2-SVM model across accuracy, precision, recall, F1-score, and AUC-ROC.

confusion matrix, which is diagonally dominant with 37 of 381 test images misclassified. Figure 4 shows predicted-confidence concentration at high values, supporting reliable operation.

B. Comparison with Existing Approaches and Backbones

Table II compares our framework with published solar dust-detection methods and with the three backbones evaluated in this work. On a binary clean/dirty protocol our EfficientNet-B2 + SVM model (90.29%, 7.77M params, multi-method XAI) is evaluated on a larger, more diverse multi-source corpus than the literature baselines (SolNet 98.2% on 2,231 images, Bassil et al. 97%), while our MobileNetV3-Large + SVM variant (88.71%, 3.00M params) offers the smallest footprint and ResNet50 + SVM matches the deployed model (90.29%, 23.59M); all three of our variants exceed Alatwi et al.'s DenseNet-169 + SVM baseline (86.79%). Reported accuracies are on respective datasets, so the comparison is indicative.

C. Cross-Dataset Generalization

To verify that the detector is not overfit to a single acquisition source, we evaluate the deployed EfficientNet-B2 + SVM model on two *fully independent* external datasets never seen during training. On the 2,562-image binary Dusty/Clean set the model reaches **98.24%** accuracy with 98.0% dirty-class recall, and on the clean/dirty subset of the 6-class faulty-panel set (383 images) it reaches **99.74%** accuracy with 100% dirty-class recall (Table III). The near-perfect external scores confirm that the frozen ImageNet representation plus the SVM head learns source-independent soiling cues rather than dataset-specific artefacts.

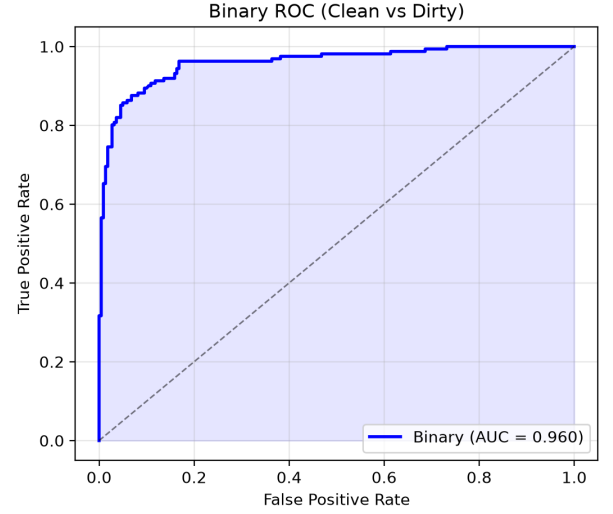


Fig. 2. Binary ROC curve for clean vs. dirty classification (AUC = 0.9596).

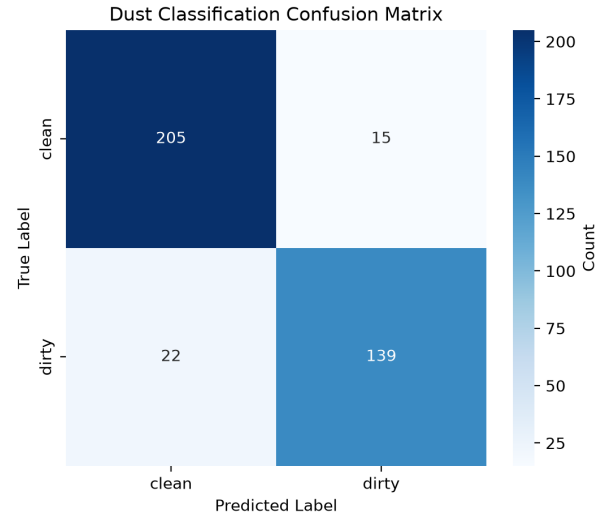


Fig. 3. Confusion matrix for the binary dust-classification task on the test set.

D. XAI Analysis

We inspect why the hybrid model reaches its decisions using the five methods.

1) *Grad-CAM Localization*: Grad-CAM heatmaps on the last convolutional block of EfficientNet-B2 weight each spatial feature map by the gradient of the dirty-class score. On clean panels activation spreads broadly across the glass surface, whereas on dirty panels it concentrates on texture-rich, low-contrast regions where soiling is visible (Figure 5).

2) *Score-CAM and Integrated Gradients*: Score-CAM produces gradient-free, high-fidelity localisations, while Integrated Gradients provides pixel-level attributions satisfying the completeness axiom, both confirming that dust-covered texture regions dominate the decision.

3) *SHAP Feature Analysis*: SHAP applied to the 1,408-dimensional feature vector shows that the top-contributing channels correspond to high-frequency texture filters and low-

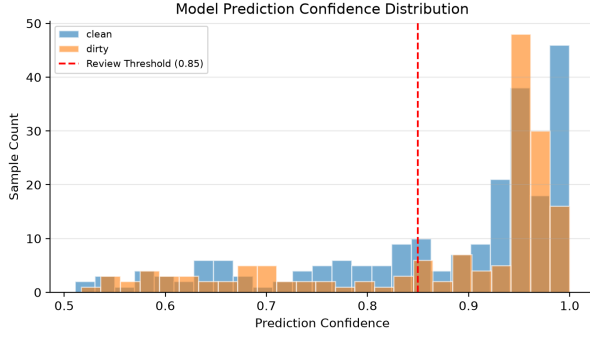


Fig. 4. Prediction-confidence distribution across test samples.

TABLE II
COMPARISON WITH EXISTING METHODS AND BACKBONE ABLATION

Method	Accuracy	Params	XAI
SolNet (custom CNN) [9]	98.2%	56M	–
DenseNet-169 + SVM [1]	86.79%	20M	–
EffB7 + ViT + tree [2]	97%	–	–
Ours (EfficientNet-B2 + SVM)	90.29%	7.77M	Multi
Ours (MobileNetV3-Large + SVM)	88.71%	3.00M	Multi
Ours (ResNet50 + SVM)	90.29%	23.59M	Multi

luminance colour channels, physically consistent with how a uniform dust layer flattens surface texture and attenuates light (Figure 6).

4) *LIME Explanations*: LIME produces sparse, localised explanations that tightly bound dust regions and are interpretable for non-technical operators in the field.

V. CONCLUSION

We presented a hybrid EfficientNet-B2 + SVM framework for binary solar-panel dust detection that achieves 90.29% test accuracy (0.9596 AUC, 86.38% CV) on a 3,787-image merged multi-source corpus with only 7.77M parameters, and generalises to two independent external datasets at 98.24% and 99.74%; MobileNetV3-Large reaches 88.71% (3.00M params) at the smallest footprint. A five-method explainability module makes every prediction auditable. Limitations include the per-source dataset sizes and the binary formulation; future work will extend to dust-severity grading, larger multi-site corpora, and TensorFlow-Lite edge deployment for on-panel inference.

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TABLE III
EXTERNAL VALIDATION ON INDEPENDENT DATASETS (DEPLOYED EFFICIENTNET-B2 + SVM)

External dataset	Accuracy	Dirty recall
Dusty/Clean (2,562 imgs)	98.24%	98.0%
Faulty-panel clean/dirty (383 imgs)	99.74%	100%

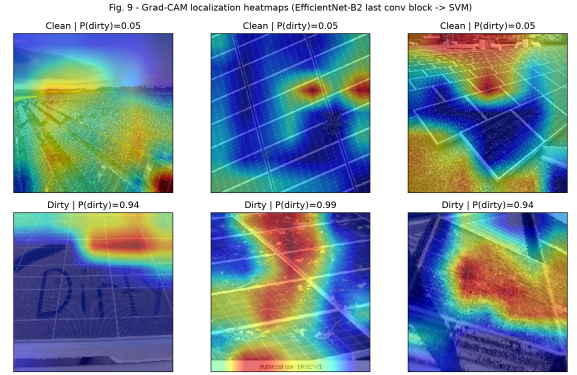


Fig. 5. Grad-CAM localisation heatmaps overlaid on clean and dirty panels, concentrating on soiling-bearing surface regions.

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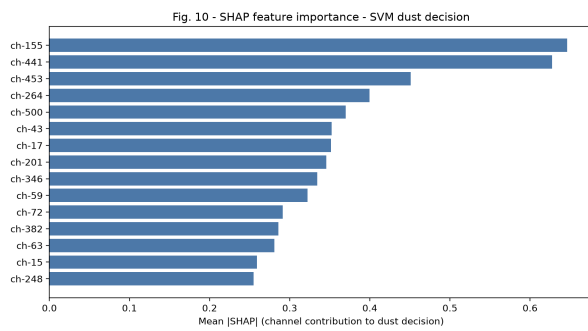


Fig. 6. SHAP feature importance (top contributing texture and colour channels) for the SVM dust decision.